AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM105
Introduction to Computer Programming

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 6/Dec. /2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 1.5 hours

Instructions:

- Total Marks: **60**
- This exam consists of 5 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Question 1 [6 Marks]

For each question, choose the most appropriate answer. Copy the answer to the External Answer booklet. Each question carries 2 marks.

1. What is the type of the variable `x` in this statement: `x = 8.8;`
   a. float
   b. int
   c. char
   d. String

2. A _____ is a program written in a high-level programming language.
   a. Source code
   b. Compiler
   c. Machine code
   d. Interpreter

3. Which of the following is NOT a Java reserved word?
   a. static
   b. void
   c. class
   d. String

Question 2 [12 marks]

The following code includes some errors. Find the errors and rewrite the code after correction.

```
public class MTA {
    public static void main(String[] args){
        int x,y;
        x=10;
        y=-10;
        if x>y and y>0
            x+=10;
            y-=10;
            System.out.println("x is:"+x);
            System.out.println("y is:"+x);
        else
            System.out.println("Condition is not applied);
    }
```

Question 3 [15 Marks]

Trace the following pieces of code and write their exact output:

a.

```
int i= 6;
int j = 3;
System.out.println(i/j);
System.out.println(i%j);
System.out.println(i/2*j+2);
System.out.println(i/2*(j+2));
```

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b.

```
int i = 0, j = 2;
if (i != j)
     if (j > 10)
          System.out.println("Condition A");
     else
          System.out.println("Condition B");
else
     System.out.println("Condition C");
     System.out.println("Condition D");
```

c.

```
int i = 8, j = 15;
if (i < j && j == 15)
   System.out.println(1);
else
   System.out.println(2);
if (i == j && i > 5) {
   System.out.println(4);
   i++;
   System.out.println(i);
} else {
   System.out.println(5);
   j++;
   System.out.println(j);
}
```

d.

```
int i;
for (i=2 ; i<=10; i*=2)
   System.out.println(i);
System.out.println(i);
```

For questions 4 and 5, copy the below code into your answer sheet and continue the rest of the code according to the given specifications of each question.

```
import java.util.Scanner;
public class MTA {
      public static void main(String[] args) {

    }
}
```

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Question 4 [20 Marks]:

Write a Java program to calculate the total internet café cost for a customer based on the number of hours used, where the usage cannot exceed 12 hours.

The program should work according to the following specifications:

- Read an integer representing the number of hours used.
- Calculate and print the total cost or a message stating that the usage duration is not allowed based on the following table:

Number of hours used	Cost per hour
Hours>12	Usage duration not allowed
hours ≤ 2	3 $ per hour
2 < hours ≤ 6	2.5 $ per hour
6 < hours ≤ 12	2 $ per hour

The total cost = number of hours used × cost per hour.
Here are different samples of the program run to help you understand how the program works:

1	Enter the number of hours: `15` Usage duration not allowed
2	Enter the number of hours: `2` The total cost is: `6$`
3	Enter the number of hours: `4` The total cost is: `10$`
4	Enter the number of hours: `10` The total cost is: `20$`

Question 5 [7 Marks]:

Write a for loop to print all odd numbers from 21 down to 1 in one line.

The output should be exactly as follows:

21 19 17 15 13 11 9 7 5 3 1

End of questions

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